

Software Requirements Specification

ResQTh

Introduction to Software Engineering (15031001) · Week 3 · Requirements & Specification

Field	This example
Team name	ILY3000
Project title	ResQTh - Intelligent emergency response and post-incident action guide for Thailand
SDLC chosen in M1 + reason	Incremental — ship the MVP first, add any additional features will be later
Date / version	v1.0 · Week 3
GitHub repo	n/a

1. Introduction

1.1 Purpose and scope

The purpose of this project is to develop an AI-assisted emergency information system within Thailand that enables users to quickly access reliable emergency resources and actionable guidance based on their location and situation. Rather than serving as a static directory, the system dynamically distinguishes between high-urgency crises (offering one-tap contact and immediate assistance cards) and low-urgency post-incident scenarios (offering step-by-step checklists and facility navigation).

The MVP scope is deliberately focused on four core functional user stories: detecting location (FR-1), filtering emergency categories (FR-2), rendering context-aware emergency tools based on urgency level (FR-3), and providing an AI-assisted natural-language search parser (FR-4). Advanced features such as offline local LLM execution, voice query support, and automated emergency service dispatches are explicitly reserved for future incremental iterations.

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
Foreign residents or visitors in Thailand	primary user	Emergency contact, information and guideline in English	Confusion or navigating to the wrong place during a crisis, not knowing what to do
Emergency Operators/ Dispatchers	secondary stakeholder	Assurance that the AI-assisted system provides only verified contacts and information from the database.	Foreigners calling the wrong department because the AI misinterpreted their request.
Technical Team/ Data Provider	institution or supplier	Clear data boundaries and reliable emergency database management	System outage during critical queries or unauthorized data manipulation and data leakage

2. Overall description

2.1 Product context — what it does / does not

ResQTh provides relevant information regarding emergency contact and detailed guidelines for users in Thailand. Everything below is the whole MVP for this term — nothing else is implied.

IN SCOPE (MVP this term)	OUT OF SCOPE (explicit promise)
• FR-1 Detect user's location • FR-2 Select an emergency category • FR-3 Display the information • FR-4 Add via Natural-language (AI)	• Group chat / messaging inside the app • Any payment or subscription feature • Display any ads-related information such as private hospitals or services • Give medical treatment instructions. • Direct automated despatching of police or medical units (the app provides one-tap phone calls and GPS sharing, but does not auto-dispatch responders)

2.2 Assumptions and constraints

- Assumption: Every user has a mobile device or browser with active internet connectivity and location services enabled.
- Constraint: One LLM API call per natural-language query; no custom model training or fine-tuning.
- Constraint: Thai and English UI text; no paid API beyond the single LLM call per request.
- Constraint: The system will use an external location/map API instead of developing its own mapping system, resulting in cost and performance constraints.

3. Functional requirements

Format: As a <role>, I want <goal>, so that <benefit>. Every story below traces to a pain observed in a (simulated) student interview, and every acceptance criterion is something a test could actually check.

FR-1 — Detect user's location

User story: As a foreign resident in Thailand, I want the system to detect my current province automatically or let me set it manually, so that I receive localized emergency contacts and facility routing tailored to my exact area.

Pain this traces to: In an emergency, users often do not know their exact province or cannot spell local Thai location names accurately under stress.

1. Given that the user has enabled location permissions, when the system receives the GPS coordinates, it identifies the current province and filters all subsequent emergency results to match that area.
2. Given that the user denies location permission or GPS signal fails, when the system loads the interface, then it prompts the user to manually select their province from a fallback menu before loading emergency data.

FR-2 — Select an emergency category

User story: As a user in distress, I want to filter emergency options by specific incident categories, so that the system immediately routes me to the right authority and appropriate urgency interface.

Pain this traces to: Panic during emergencies makes reading through unstructured lists difficult, while choosing the wrong department causes dangerous delays in getting help.

1. Given a list of predefined category options (e.g Medical Emergency, Police/Crime, Tourist Police, Lost Passport), when the user selects a category, then the system filters contacts matching that category and assigns its corresponding urgency tag (High-urgency or Low-Urgency) .
2. Given an active category selection, when the user clears the category filter, then the system resets the view to display general emergency hotlines for the selected province.

FR-3 —Display the information

User story: As a user in Thailand, I want the system to present contact details and action tools tailored to the urgency of my situation so that I can take immediate action during a crisis or follow a structured checklist post-incident.

Pain this traces to: Users struggle to navigate static lists during panic, whereas post-incident scenarios require procedural guidance rather than just phone numbers.

1. Given a selected or parsed emergency category tagged as high-urgency (e.g., active crime, medical emergency), when the system renders the emergency information, then it displays a high-urgency interface featuring one-tap hotline dialing, a high-contrast localized assistance card, and quick GPS coordinate sharing.
2. Given a selected or parsed emergency category tagged as low-urgency (e.g., lost passport, non-violent theft), when the system renders the emergency information, then it displays a low-urgency interface featuring an interactive step-by-step resolution checklist, incident note logging, and routing to official facilities.
3. Given a scenario where no local contact matches the exact category in that province, when the system executes the search, then it displays the national emergency hotline as a default fallback.

FR-4 — Add via Natural-language (AI-assisted)

User story: As a foreigner, I want to type a query like "I lost my passport in Phuket" or "car accident near Chiang Mai", so that the system automatically identifies the appropriate emergency category and location, and urgency level.

Pain this traces to: Foreigners do not know local emergency classification terms or which specific department handles their scenario

1. Given a natural-language emergency prompt, when the user submits it, then the system extracts the category, province, and urgency level, automatically redirecting the user to the matching FR-3 display interface.
2. Given an incorrect AI suggestion, when the user manually overrides the category or province draft before searching, then the system updates results using the user's manual selection.

4. Non-functional requirements

Forbidden words unless quantified: fast, easy, user-friendly, smart, secure. NFR-5 (security/privacy) is required for M2 — every SRS must have at least one.

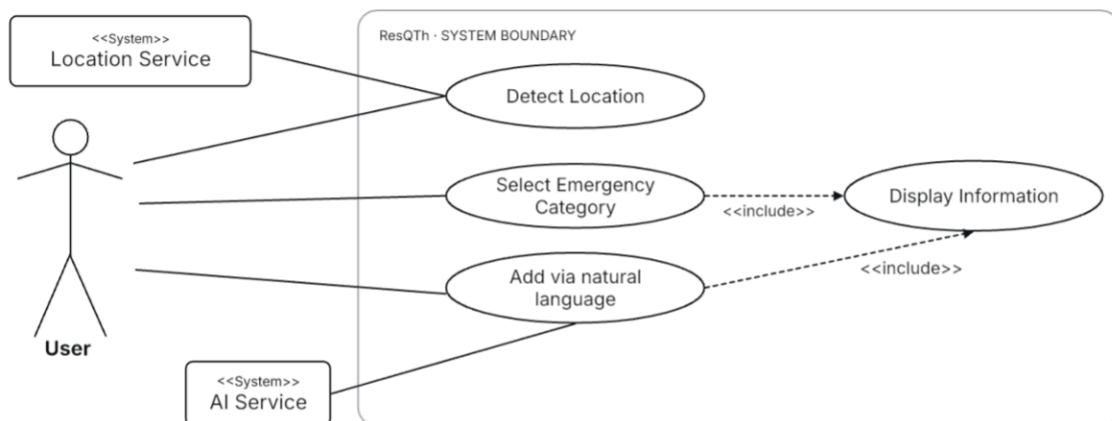
ID	Kind	Statement (testable)	How we will check
NFR-1	performance	Emergency action interfaces(high-urgency one-tap view or low-urgency checklist view shall render in under 3 seconds over standard internet or wifi	Stopwatch on a laptop, 3 trials.
NFR-2	performance	The natural-language AI feature shall parse prompt text, identify location/category/urgency-level and redirect to the display interface within 4 seconds of submission.	Measure API response time using network logs over 10 test queries.
NFR-3	usability	A first-time user shall find the relevant emergency hotline in under 20 seconds without asking a teammate.	Timed trial with a user who has not seen the app interface.
NFR-4	reliability	The offline database cache shall display pre-loaded national emergency hotlines and basic step-by-step action checklists for common low-urgency categories if network connectivity fails.	Disconnect device network connection and verify both hotline numbers and checklist steps remain visible
NFR-5	security / privacy (required)	The system shall not store or transmit user location coordinates or raw text queries tied to personal identifying data.	Audit server logs and browser local storage during search requests.

5. Use cases

5.1 Use-case list

Use case	Actor	Related FR	Brief success story
UC-1 Detect location	User	FR-1	User location identified as “Phuket”; system displays local emergency directory
UC-2 Filter category	User	FR-2	User selects “Tourist Police”; the system filters out non-relevant hotlines.
UC-3 Display emergency information	User	FR-3	The system detects urgency level and renders either high-urgency tools (one-tap hotline call, Thai translation card, GPS broadcast) or low-urgency tools (step-by-step checklist, facility routing).
UC-4 Add via Natural-language	User	FR-4	User types <i>"stolen bag in Bangkok"</i> ; system extracts category, location, and low urgency level, automatically redirecting to the FR-3 display interface.

5.2 Use-case diagram



6. Out of scope

1. Group chat or messaging inside the app.
2. Any payment or subscription feature.
3. Display any ads-related information such as private hospitals or services
4. Give medical treatment instruction
5. Direct automated dispatching of police or medical units (the app provides one-tap phone calls and GPS sharing, but does not auto-dispatch responders)

7. Traceability matrix (Golden Thread)

Only FR-3 is filled all the way across — that is the single thread this course follows week by week (Ch3 → Ch4/5 → Ch6 → Ch7). The other rows stay open until their week arrives, exactly like your own M2 should.

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Unfamiliarity with regional hotline numbers	FR-1 detect location	(Week 5)	(Week 6–7)	(Week 7)
Confusion selecting the right department	FR-2 select category	(Week 5)	(Week 6–7)	(Week 7)
Panic during high-risk events/ lack of post-incidence guidance	FR-3 Display the information	“Emergency Directory” UI component(Ch6)	(Week6-7)(Ch6)	get_emergency_info() unit tests(Ch7)
Foreigner unaware of local terms	FR-4 Add via NL	(Week 5)	(Week 6–7)	(Week 7)

8. AI usage log

If this table is empty but the prose looks generated, the milestone is capped at 50% — log every prompt that drafted a story or paragraph, honestly.

Date	Tool model /	What we asked	What AI produced	What we changed / verified	What we learned
18 Aug 2026	Gemini	Emergency information website for both locals and foreigners that has any name that is easy to remember and shorter	. safebridge . ResQ	Added “Th” at the end of “ResQ” as we wanted to represent our services in Thailand	
23 Aug 2026	Gemini	Draft our scattered thoughts and ideas into structured sections that align with the provided project layout. (we provided our ideas)	Updated SRS document sections matching exact course structure.	-Edit stakeholder, Technical team need and what they fear -Updated FR, NFR section to align with our project scope -Update the constraints of our concern -Edit use case of our preference	-how to layout our requirements and preferences in our project -how to use Ai to draft our ideas and modify it according to our project requirements and do the review
4 Sep	ChatGPT	Check any typo errors in our documents	Parts where I am having typo errors	-We scanned the document again and check pointed areas if the errors were really there and fixed what was needed	how to use AI as a tool on time-consuming and tedious tasks, and we verify the results as humans

9. Self-check

- ✓ Every FR is a user story with a benefit — all FRs in §3 end in a “so that” clause tied to a real pain.
- ✓ Every FR has acceptance criteria you could actually test — 2 AC per FR, each a number, a state, or a yes/no.
- ✓ No FR is vague — no bare “fast”, “easy”, or “user-friendly” anywhere in §3.
- ✓ NFRs exist, including security / privacy — NFR-5 covers the required privacy check.
- ✓ Every FR traces back to a stated problem — see “Pain this traces to” under each FR, and §7.
- ✓ Scope and out-of-scope are stated — §2.1 and §6.
- ✓ Use-case list matches the FR list — 4 use cases, one per FR (diagram added Week 4).
- ✓ AI log is honest — §8 shows exactly what was asked, drafted, and hand-corrected.

Course rule: you may use AI to draft. You must log it, verify it against a real stakeholder, and be able to explain every FR in the oral exam. Work you cannot explain earns no credit.